Southern New Hampshire University

CS 255 Business Requirements Document

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design a web-based system for DriverPass that helps customers prepare for DMV driving tests through online classes, practice tests, and on-the-road training.
- The system will give customers a way to manage their training and appointments while giving DriverPass employees the tools they need to manage customers, reservations, packages, and system access.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass was created because many people are not fully prepared for their DMV driving tests. The company wants to provide more complete training instead of having students rely only on previous test questions.
- The system will support online classes, practice tests, driving lesson reservations, customer information, driver notes, and training packages.
- Customers will be able to schedule, cancel, or change driving appointments online. The secretary will also be able to make appointments for customers who call or visit the office.
- The system will track the customer, driver, car, date, and time connected to each driving lesson.
- DriverPass also needs the system to receive DMV updates so its rules, policies, and practice material stay current.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Give customers access to online training materials and practice tests.
- Allow customers and authorized employees to schedule, modify, and cancel two-hour driving lesson appointments.

- Track each reservation with the correct customer, driver, vehicle, date, and time.
- Show practice-test progress, including the test name, time taken, score, and status.
- Allow DriverPass to offer its three training packages and disable a package when it is no longer available.
- Maintain different access levels for the owner, IT officer, secretary, drivers, and customers.
- Keep an activity history showing who created, changed, or canceled reservations.
- Receive notifications when the DMV provides updated rules, policies, or sample questions.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system will operate as a web-based, preferably cloud-hosted system that customers and employees can access through an internet connection.
- The system should provide dependable access from computers and mobile devices so users can complete normal tasks without unnecessary delays.
- DMV-related training information should be updated when DriverPass receives notice of new rules, policies, or sample questions.
- Reports and selected information should be downloadable so authorized users can work with the information in programs such as Excel.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The main system will run through a web browser so DriverPass is not limited to one desktop operating system.
- The interface must work on computers and mobile devices.

- The system will require back-end data storage for customer accounts, reservations, packages, test results, driver notes, vehicles, and activity records.
- Because DriverPass prefers a cloud-based system, backup and security services should be handled by the hosted environment rather than requiring DriverPass to maintain them locally.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Each user will have an individual account so the system can identify who performs an action.
- Reservations must accurately connect the customer with the correct driver, vehicle, date, and time.
- The activity history must record who created, modified, or canceled a reservation so DriverPass can review responsibility when a problem occurs.
- Practice-test records must accurately show the test name, time taken, score, and status as not taken, in progress, failed, or passed.
- Customer records should keep required information such as name, address, phone number, pickup location, and drop-off location associated with the correct customer.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The IT officer will have full administrative access to maintain accounts, reset access, and block an employee account when needed.
- Authorized users should be manageable through account and role settings without requiring changes to the main program code for routine account changes.
- The system must allow DriverPass to disable an existing training package when the company no longer wants customers to register for it.
- The design should allow future development to add or remove package modules, although the interview states that this does not need to be part of the first release.
- A web-based design will make it easier to support updated browsers and devices without installing a separate application on every user's computer.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users will log in with individual credentials, and access will be based on their assigned role and responsibilities.
- The IT officer will have full account access, including the ability to reset accounts and block access for employees who should no longer use the system.
- Customers who forget their passwords will be able to use an automatic password-reset process.
- Sensitive customer and payment information must be protected when it is stored and exchanged between the user's device and the system.
- The system should limit repeated failed login attempts to reduce the risk of unauthorized access and alert or require additional account verification when suspicious activity occurs.
- The system will maintain an activity record so DriverPass can identify who created, changed, or canceled reservations.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow customers to create and use an account to access DriverPass services online.
- The system shall allow authorized users to enter and maintain customer information.
- The system shall provide customers with online classes and practice tests based on their training package.
- The system shall record practice-test progress, including test name, time taken, score, and status.
- The system shall allow customers to schedule, modify, and cancel driving lesson appointments online.
- The system shall allow the secretary to schedule appointments for customers who call or visit the office.

- The system shall schedule driving lessons in two-hour sessions.
- The system shall associate each driving reservation with the correct customer, driver, vehicle, date, and time.
- The system shall support DriverPass training packages and allow an authorized user to disable a package.
- The system shall store driver notes and lesson start and end times.
- The system shall allow authorized users to download reports and selected information for use outside the system.
- The system shall record who created, modified, or canceled a reservation and allow an activity report to be produced.
- The system shall allow customers to automatically reset a forgotten password.
- The system shall receive or support updates from the DMV for new rules, policies, and sample questions and notify DriverPass when updates are available.
- The system shall provide a way for customers to contact DriverPass and for DriverPass to contact students.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface will be web-based and accessible through a browser on computers and mobile devices.
- Customers will use the interface to view online classes and practice tests, check test progress, manage appointments, and reset forgotten passwords.
- The secretary will use the interface to enter customer information and create, change, or cancel appointments.
- The IT officer will use administrative controls to maintain the system and manage user accounts and access.
- The owner will need access to business information, reports, reservation activity, customer progress, and driver notes.
- Customer information pages will include fields such as first name, last name, address, phone number, pickup location, and drop-off location.

- The progress area will show the test name, time taken, score, and status. Driver information will show lesson times and driver comments.
- The interface will also provide a contact page and a way for DriverPass to contact students.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Customers and employees who use the online system will have access to a compatible web browser and an internet connection.
- Customers will provide accurate registration, contact, pickup, drop-off, and payment information.
- DriverPass will provide authorized staff with the correct roles and access levels.
- The DMV will make updated rules, policies, or sample questions available in a form that DriverPass can receive or use.
- Cloud hosting will provide the backup and security support requested by DriverPass.
- Routine package disabling is included in the first release, while adding or removing package modules will require future developer or systems analyst work.

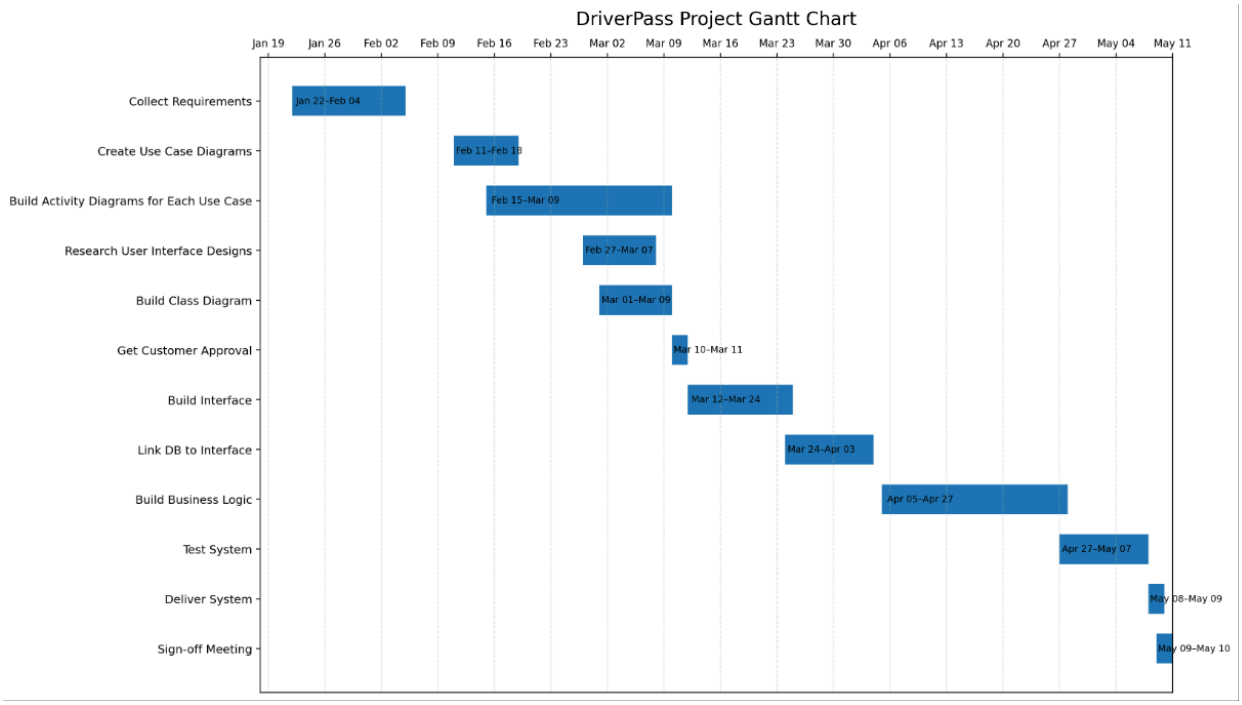
Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Data can be viewed or downloaded for offline work, but records cannot be modified or updated offline because that could create duplicate or inconsistent data.
- The first release will not allow a nondeveloper to add or remove package modules. That capability may be considered in a future release.
- The system depends on internet access for online account changes, reservations, classes, practice tests, and other live services.
- DMV content can only stay current if DriverPass continues to receive usable updates from the DMV.
- The project schedule is limited by task dependencies and staff availability, including the planned start date for the class diagram.
- Customer-requested changes after the approval meeting could affect the planned development and delivery dates.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.



Gantt chart draft based on the dates provided in the DriverPass interview transcript.